What Survives the Ghost Test?

A Four-Level Classification in Pure-Weyl Gravity

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23 July 2026

Abstract

Fourth-order gravity predicts more gravitational-wave solutions than general relativity. Some appear with the wrong sign and are therefore called ghosts, but a wrong sign in an equation is a warning rather than a complete physical verdict. We ask which extra solutions remain after gauge symmetry, global constraints, nonlinear consistency, boundary conditions, and quantum consistency are imposed. Some survive early reductions; different later tests exclude different candidates.

Pure Weyl gravity is a fourth-order conformal metric theory. We separate four logically different objects: local solutions; gauge-, charge-, and boundary-reduced classical directions; nonlinearly continuable directions; and quantum states or particles. This separation is also corrective: a boundary conclusion valid for an incomplete reconstruction need not survive reconstruction of the metric and its symplectic current.

On a compact Weyl–Maxwell laboratory, additional axial and polar directions survive local gauge reduction and have nonzero induced Lee–Wald pairings; the associated positive-frequency Hermitian signs do not follow the simple rule "Einstein positive, additional negative." At second order, finite-harmonic exponential-polynomial continuation is equivalent to five quadratic stabilizer-charge conditions, whereas bounded continuation obeys further resonance constraints. On Schwarzschild, restoring omitted metric-reconstruction equations reverses an earlier Einstein-selection interpretation. In the complete axial $\ell = 2$, $M = 1$, $\omega \in [1/2, 3/4]$ pilot, the missing row propagates as $C' = -2C/r$ and its zero fibre is a six-dimensional formal module. The repaired non-Einstein X_0 direction is finite in the fixed representative and under rate-zero Einstein shears, but crosses true oscillatory Einstein shears divergently. Separately, the generic-angular polar module has a mixed finite line away from an exact algebraic exceptional locus. These are formal-infinity results, not an asymptotically flat scattering theorem. A separate exact Phase-3A endpoint construction gives three-dimensional L^2 wave-packet trace spaces at both axial null endpoints on the same pilot interval. Their action-derived flux Grams have rank three, zero radical, and inertia $(1, 2, 0)$ for $\alpha_{\mathrm{W}} > 0$, with no frequency wall. This endpoint module has an exact triangular filtration into two spin-two Regge–Wheeler factors and one spin-one Regge–Wheeler factor. Their real-potential Wronskians prove that the map from future-horizon-regular data to the incoming $\mathscr{I}^-$ trace is invertible for every real $\omega > 0$. Thus the full indefinite incoming endpoint space is globally populated. Its metric/carrier spin-two extension is hyperbolic, while the orthogonal spin-one quotient lift is strictly negative. The outgoing reflection block, damped poles, time-domain PDE stability, CPT positivity, and a complete scattering operator remain open. The exact filtration does, however, exclude exponentially growing axial $\ell = 2$ separated modes.

Strict fixed-field-content pure Weyl gravity has the established nonzero local Euclidean one-loop Weyl-anomaly coefficients, here reconstructed from the declared determinant complex

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and mapped into local BV cohomology. A formal compensator makes the cocycle exact but changes the theory. On selected compact reduced blocks, an exact pilot classifies a nonempty 24-real-dimensional cone of symmetry-compatible positive pseudo-Hermitian metrics. It does not construct Mannheim's CPT prescription: independent parity/time data and BRST descent are absent or obstructed. A changed two-phase counterflow theory supplies a complete free causal BV complex at a specified Berger background, but its physical $j = \frac{1}{2}$ block has a family-persistent Hamiltonian–Hopf quartet. Its complex spectrum forbids a positive pseudo-Hermitian rescue. Thus pole signs are too coarse, but no candidate studied here reaches a positive full-BV state, scattering theory, or unitarity theorem.

Contents

1 The question and the four levels

The metric action of strict pure Weyl gravity is

$$S_{\mathrm{W}}[g] = \alpha_{\mathrm{W}} \int_M \sqrt{-g}\, C_{\mu\nu\rho\sigma} C^{\mu\nu\rho\sigma}\, \mathrm{d}^4 x, \qquad B_{\mu\nu} = 0, \tag{1}$$

where $B_{\mu\nu}$ is the Bach tensor. Around a fixed background, the linearized Bach equation is fourth order. A formal factorization, a partial fraction decomposition, or a negative residue can reveal an indefinite direction. None of these alone says whether the direction survives gauge reduction, global constraints, physical boundary conditions, nonlinear integrability, or the construction of a quantum state.

This paper organizes the first phase of a computer-assisted research programme by the following hierarchy:

$$\boxed{\begin{array}{ll} \text{Level 1:} & \text{local solutions of the linear field equations,} \\ \text{Level 2:} & \text{reduced classical directions after gauge, charge, and boundary conditions,} \\ \text{Level 3:} & \text{directions tangent to nonlinear solutions in a stated correction class,} \\ \text{Level 4:} & \text{quantum states, particles, and a positive physical probability interpretation.} \end{array}} \tag{2}$$

The arrows between these levels are mathematical problems. They are not terminological promotions. A nonzero classical symplectic pairing at Level 2 is not a positive quantum norm at Level 4. A secular second-order primitive at Level 3 is not nonlinear stability. A local Euclidean anomaly calculation is not a Lorentzian scattering theorem.

The phase-one question is therefore not simply "does the propagator contain a ghost?" It is:

> Which additional fourth-order directions survive each declared reduction, which fail nonlinear or boundary tests, and what remains undecided before a particle interpretation can even be posed?

The answer is mixed. Some additional directions survive local gauge and presymplectic reduction. Compact nonlinear constraints remove some of them, while balanced combinations can continue formally with secular terms. A complete axial $\ell = 2$ calculation finds a finite formal additional direction but no representative-independent quotient statement, while the generic-angular polar calculation finds mixed finite radial directions. A structured positive pseudo-Hermitian metric exists on selected finite reduced blocks, but no genuine BRST-compatible C operator or full-BV positive state has been constructed. The strict quantum gauge theory is locally anomalous. Finally, a changed theory that passes a difficult causal construction fails a physical spectral test that no choice of positive inner product can repair. Different candidates fail at different levels.

The framework also corrected one of the programme's own conclusions. At $\ell = 2$ and $M\omega = 3/5$, an earlier Schwarzschild reconstruction appeared to make the additional axial direction radially divergent. A first audit restored differentiated forcing, but the Phase-3 audit then found that the independent $v\varphi$ Ricci equation had still been omitted. Its defect obeys $C' = -2C/r$. On the complete zero fibre, the repaired X_0 direction is finite in the fixed representative and under the rate-zero Einstein kernel, whereas a true oscillatory Einstein shear produces a divergent cross term. The superseded tables and the former unrestricted representative-independence claim are retained as failed promotions rather than used as evidence for Einstein selection.

2 Theories, conventions, and scope

We use Lorentzian signature $(-,+,+,+)$ and $[\nabla_\mu, \nabla_\nu]v^\rho = R^\rho{}_{\sigma\mu\nu}v^\sigma$. Classical fields are real; complex Fourier modes are bookkeeping devices and physical pairings use the corresponding real or conjugate-frequency form. The connected local gauge group of the strict theory is

$$\mathrm{Diff}_0(M) \ltimes C^\infty(M, \mathbb{R})_{\mathrm{Weyl}}, \qquad g_{\mu\nu} \mapsto e^{2\sigma} g_{\mu\nu}. \tag{3}$$

Unless a theorem card says otherwise, no boundary counterterm or topological density is included in the action. Boundary and support conditions are part of the theorem, not implicit conventions.

Two comparison theories appear. On the compact product we use

$$S_{\mathrm{WM}}[g,A] = S_{\mathrm{W}}[g] - \frac{1}{4}\int_M \sqrt{-g}\, F_{\mu\nu}F^{\mu\nu}\,\mathrm{d}^4x, \tag{4}$$

$$S_{\mathrm{EM}}[g,A] = \frac{1}{2\kappa}\int_M \sqrt{-g}\,(R-2\Lambda)\,\mathrm{d}^4x - \frac{1}{4}\int_M \sqrt{-g}\, F_{\mu\nu}F^{\mu\nu}\,\mathrm{d}^4x. \tag{5}$$

The magnetic bundle is fixed in that comparison. Einstein–Maxwell is a source theory and comparator; it is not identified with a subsector carrying the same symplectic structure.

The quantum repair uses a formal Wess–Zumino field τ with $s\tau = \omega$, where ω is the Weyl ghost, and the dressed metric $\widehat{g} = e^{-2\tau}g$. This is an enlarged BV theory. It is not strict pure Weyl gravity in another gauge.

The later counterflow experiment changes the classical action as well. Its defining matter mechanism is two positive phase fields with an auxiliary diagonal $U(1)$ connection,

$$S_{\mathrm{phase}} = -\frac{1}{2}\int_M \sqrt{-\widehat{g}}\,\left[f_1^2(\mathrm{d}\theta_1 - A)^2 + f_2^2(\mathrm{d}\theta_2 - A)^2\right]. \tag{6}$$

Writing

$$\chi = \frac{f_1^2\theta_1 + f_2^2\theta_2}{f_1^2+f_2^2}, \qquad \psi = \theta_1 - \theta_2, \qquad \mu^2 = \frac{f_1^2 f_2^2}{f_1^2+f_2^2}, \tag{7}$$

separates a diagonal gauge phase from a gauge-invariant relative phase with positive bare kinetic coefficient μ^2. Fixing the harmless positive diagonal normalization to $f_1^2+f_2^2 = 4$, the complete counterflow action used below is

$$S_{\mathrm{cf}} = \int_M \sqrt{-\widehat{g}}\left[\frac{\alpha_B}{8}\widehat{C}_{\mu\nu\rho\sigma}\widehat{C}^{\mu\nu\rho\sigma} + \frac{M_2}{2}\widehat{R} - V_0 - \frac{\mu^2}{2}(\widehat{\nabla}\psi)^2 - 2(A-\mathrm{d}\chi)^2\right]\mathrm{d}^4x. \tag{8}$$

There is no Maxwell kinetic term for A. Besides diffeomorphisms, the local gauge transformations are

$$g \mapsto e^{2\sigma}g, \quad \tau \mapsto \tau+\sigma, \qquad \chi \mapsto \chi + \lambda, \quad A \mapsto A + \mathrm{d}\lambda, \tag{9}$$

so that $\widehat{g}$ and ψ are gauge invariant. Constant shifts of ψ generate the physical relative symmetry R_{rel} rather than a local gauge symmetry. Equation (8), its BV cotangent lift, and ordinary nonminimal gauge-fixing doublets define the changed theory; a component Hessian is evidence for this action, not its definition. Results for the changed theory are never imported back into the strict theory.

2.1 The scope tuple

The mathematical scope of every result is typed by

$$\boxed{\mathfrak{S} = (\mathcal{T}, M, \bar{\Phi}, \mathcal{G}, \mathcal{X}, \mathcal{B}),} \tag{10}$$

where the entries are the theory, spacetime, background fields, gauge group, function or phase space, and boundary/support class. Evidence is recorded separately as

$$\mathfrak{E} = (\text{claim identifier}, \text{dependency tags}, \text{evidence state}, \text{limitation}). \tag{11}$$

Thus an epistemic label is never treated as part of the mathematical object. Two statements with different entries in $\mathfrak{S}$ are not combined without a constructed map. This matters because the programme uses several controlled laboratories:

Laboratory	Question isolated there
$\mathbb{R} \times S^3$ conformal cylinder	Complete free causal complex, a selected residual zero-charge reduction, and a PT/CPT chain-map test
Biaxial Berger spacetime	Clocks, relational observables, interactions, and the counterflow successor
$\mathbb{R} \times S^1 \times S^2$ magnetic product	Einstein/additional branches, Lee–Wald pairing, Taub constraints, resonances, and a finite-block pseudo-Hermitian pilot
Nariai	Causal transfer and a scoped factor-residue sign mechanism
Schwarzschild and static spherical metrics	Horizons, formal radial-pairing filtrations, and black-hole perturbations
Euclidean S^4 and regular local charts	One-loop determinants and local BV anomaly classes

These are related tests of a common architecture, not successive approximations to one already unified universe.

3 What "ghost" can mean

The literature uses the word *ghost* for inequivalent failures. The distinctions are especially important in a gauge theory with a fourth-order equation. A modern review of the nondegenerate higher-derivative Ostrogradsky theorem is Ref. [6]; gauge degeneracy and physical reduction must be analyzed separately.

Diagnosis	Mathematical question	Phase-1 disposition
Ostrogradsky direction	Is a nondegenerate higher-derivative Hamiltonian unbounded before constraints?	Crosswalks are computed on selected blocks; this is not used as a complete physical quotient.
Negative pole residue	Does a gauge-fixed propagator decompose with a negative coefficient?	Under the standard adjoint it is an initial warning, not a final state classification. A PT/CPT prescription changes the physical-adjoint question and must pass its own structural gates.
Negative classical energy or flux	What is the sign of a declared canonical-energy form, flux, or positive-frequency Hermitian current derived from the real Lee–Wald form?	Computed for selected compact and black-hole sectors. Some signs contradict a naive Einstein/additional split.
Radical or pure gauge	Does the induced presymplectic form vanish on the direction?	Additional compact-product directions are nonradical on the declared pre-residual quotient.
Nonlinear obstruction	Is a linear direction tangent to a second-order solution in a stated function class?	Classified completely on a finite harmonic exponential-polynomial space; bounded continuation is stricter.
Negative quantum norm	Is there a BRST-compatible positive state on the full physical algebra?	Not established for the standard adjoint. Under a PT/CPT prescription, positive η alone is insufficient: genuine C, BRST descent, and causal covariance remain unconstructed.
Failure of unitarity	Does an asymptotic state space satisfy the optical theorem?	Not tested: no complete scattering or particle theory exists.
Alternative pole prescription	Does a structured positive metric, a genuine PT/CPT involution, and BRST descent exist in the declared representation?	Tested on selected reduced blocks. Positive pseudo-Hermitian metrics exist in a compact pilot, but no genuine Mannheim C operator or full-BV descent is established; the counterflow quartet is spectrally excluded.

PT-symmetric treatments of the Pais–Uhlenbeck system, Dirac–Pauli formulations of quadratic gravity, and fakeon prescriptions all demonstrate that "negative residue implies negative-probability asymptotic particle" is not the only proposed logical route [10, 16, 17]. They also demonstrate why the chosen state space and prescription must be declared. The scoped Phase-2 pilot therefore separates four gates that are often conflated: existence of a positive metric η, existence of an independent parity/time involution and corresponding C, compatibility with the BRST differential, and compatibility with a causal covariance. Passing the first gate on a finite block does not imply the other three. Fakeon and Dirac–Pauli prescriptions remain neighboring programmes rather than consequences of these reductions. The perturbative renormalizability and extra massive-spin-two spectrum that made this question central to higher-derivative gravity go back to Stelle's foundational analyses[7, 8].

4 Which additional linear directions survive reduction?

The compact magnetic product is the cleanest place to compare the Einstein–Maxwell and Weyl–Maxwell linear solution spaces. The computation uses both parities, generic harmonics, exceptional modes, and global blocks.

The sign-bearing object must be distinguished from the underlying presymplectic form. Let Ω_{WM} be the real, antisymmetric Lee–Wald form obtained from the Weyl–Maxwell action [1, 2]. Rank and radical are defined on the real reduced solution space; Ω_{WM} itself has no inertia. After complexifying, fix a common stationary shell

$$e^{-i\omega t}, \qquad \omega > 0, \qquad k = \frac{2\pi n}{L}, \qquad \lambda = \ell(\ell+1), \quad \ell \geq 2, \tag{12}$$

and impose the conjugation reality condition between positive- and negative-frequency coefficients. Here L is the circumference of S^1 and $N_{\ell m} = \int_{S^2} |Y_{\ell m}|^2 \mathrm{d}\Omega > 0$ is the chosen spherical-harmonic normalization. On the positive-frequency space define

$$h_+(u, v) = \frac{\Omega_{\mathrm{WM}}(\bar{u}, v)}{-i\omega L N_{\ell m}}, \qquad L N_{\ell m} > 0. \tag{13}$$

Reality and antisymmetry of Ω_{WM} make h_+ Hermitian. A positive change of the circle or harmonic normalization, and a nonzero complex rescaling of any basis vector, act by Hermitian congruence and leave its inertia invariant. The signatures below refer only to this h_+. Exceptional dipoles, generalized zero modes, and $\omega = 0$ are classified in separate real blocks and are not assigned these frequency-shell signatures.

Result card A: compact Einstein/additional decomposition

Theory. Einstein–Maxwell source and Weyl–Maxwell target.

Background. The magnetically supported compactified Plebanski–Hacyan product $\mathbb{R} \times S^1 \times S^2$, in a fixed magnetic bundle.

Domain. Parity-complete generic linear modes, together with the declared exceptional and global blocks.

Gauge and quotient. Local gauge quotient and maximal pre-residual H^0 exact sequence; final stabilizer reduction is not included.

Boundary condition. Spatial periodicity; no asymptotic scattering condition.

Statement. The Einstein–Maxwell solutions inject into the Weyl–Maxwell solutions. The cofiber contains additional axial and polar blocks. On each generic additional real block Ω_{WM} has zero radical and rank four. On the associated positive-frequency complex two-plane, h_+ has inertia

$(2,0)$; its restriction to the target Einstein image has inertia $(1,1)$ and the complete generic target has inertia $(3,1)$. The independently normalized Einstein–Maxwell source Hermitian current form has inertia $(2,0)$, so inclusion does not define a cyclic or symplectic equivalence.

Not claimed. No positive energy, particle norm, final residual quotient, causal boundary admissibility, or nonlinear closure follows.

The important conclusion is not that the additional block is "healthy" because the displayed frequency form is positive. The conclusion is that it is not an algebraic duplicate or a radical direction on this quotient, and that the standard pairing does not respect the naive expectation that the Einstein image should inherit the ordinary Einstein sign while the complement should carry the bad sign. The action matters: the Einstein–Hilbert and Weyl Lee–Wald forms are different even on the same metric perturbation. Neither the rank statement nor the inertia of h_+ is by itself a positive-energy or positive-norm quantum theorem. The inertia statements are also relative to the declared conjugation and frequency shell that define h_+ in Eq. (13); a different admissible adjoint would organize the same real symplectic data differently. The failure of the naive split is therefore adjoint-dependent evidence that motivates, but does not prove, alternative inner-product prescriptions of the PT/CPT type[10].

The Phase-2 finite-block pilot makes this last qualification exact. Let G denote the imported action-derived Krein form, and let H be the positive-frequency generator. In the normal spectral frame,

$$\mathcal{V} = \mathcal{V}_- \oplus \mathcal{V}_+ \oplus \mathcal{V}_p, \qquad \dim_{\mathbb{C}}(\mathcal{V}_-, \mathcal{V}_+, \mathcal{V}_p) = (2, 2, 4), \tag{14}$$

with

$$H^2 = \mathrm{diag}(q_- I_2, q_+ I_2, p I_4), \quad q_\pm = k^2 + \lambda \pm \sqrt{2\lambda}, \quad p = k^2 + \lambda - \frac{2}{3}. \tag{15}$$

The connected orientation-preserving product symmetry has exact commutant

$$\mathrm{Comm}(H) = M_2(\mathbb{C}) \oplus M_2(\mathbb{C}) \oplus M_4(\mathbb{C}), \tag{16}$$

of complex dimension 24. The complete structured Hermitian solution of the intertwining equation

$$H^\dagger \eta = \eta H, \qquad \eta = \eta^\dagger > 0, \tag{17}$$

is

$$\eta = A_- \oplus A_+ \oplus A_p, \qquad A_- \in \mathrm{Herm}_2^+, \qquad A_+ \in \mathrm{Herm}_2^+, \qquad A_p \in \mathrm{Herm}_4^+, \tag{18}$$

with rational or polynomial parameter dependence, no poles on the physical domain, and reality condition $\eta(\lambda, -k) = \overline{\eta(\lambda, k)}$. This is a $4 + 4 + 16 = 24$ real-dimensional open cone at fixed (λ, k). Positivity is equivalently the positivity of the exact LDL pivots of all three blocks; one explicit exact member is $A_- = I_2$, $A_+ = I_2$, $A_p = I_4$.

The form G is negative definite on $\mathcal{V}_-$ and positive definite on $\mathcal{V}_+ \oplus \mathcal{V}_p$. Consequently

$$C_0 = -I_2 \oplus I_2 \oplus I_4, \qquad C_0^2 = 1, \qquad \eta_0 = GC_0 > 0, \tag{19}$$

and C_0 is the unique G-self-adjoint involution in the declared commutant for which GC_0 is positive. Equations (18)–(19) transform by congruence under a change of multiplicity frame, so the cone and its positivity do not depend on the printed basis.

This is an exact finite-block quasi-Hermiticity theorem in the sense of Mostafazadeh, not yet Mannheim's more specific CPT construction. The fixed magnetic carrier supplies neither an independently declared linear parity P nor an antilinear time reversal T; orientation reversal may change the magnetic bundle sector. Thus no relation between η, P, and a physical C can be asserted, and quotient descent is not established. A representation-independent Mannheim gate would instead supply an antilinear involution Θ with $\Theta^2 = 1$ and $[\Theta, H] = 0$, together with a linear involution C satisfying $C^2 = 1$, $[C, H] = [C, \Theta] = 0$, an explicit convention relating C, Θ, G and η, and a chain map on the BRST complex. None of that missing structure is manufactured from η alone. On each chirality of the selected conformal-cylinder energy-five regression, the stationary C_0 has rank-32 commutator with every one of the eight proper-conformal rows and degree-zero-to-one BRST defect rank 102 (rank 204 on both chiralities). The all-energy representation argument reduces every residual-invariant Hermitian involution to a scalar on each chirality; neither scalar choice makes GC positive. A nontrivial action on the ghost normalizer lies outside this classified commutant and remains open. These conclusions carry the dependency tags `LOCAL-ALGEBRAIC` and `REDUCED-MODE`; they do not establish a Lorentzian state.

A second Phase-2 calculation isolates an operator mechanism on Nariai. The spin-two operator factors as $L_E L_C$ with $L_C = L_E - 2/3$, and the two factor projectors have Lee–Wald residues $-1/3$ and $+1/3$. This predicts a relative sign exchange between the two factors and is robust when both static-patch horizons are retained. It does not fix an absolute energy sign, inertia, or positivity, because Nariai has no declared global timelike generator for that purpose; nor does it identify the factors with the compact magnetic-product Einstein image and additional cofiber. It is a mechanism-level comparison across different scope tuples, not a family theorem.

The failed dyonic-family extension is recorded separately in Appendix A; it supplies no sign-family theorem.

This exact-sequence formulation also prevents an unjustified direct-sum claim. A lift of an additional curvature representative may be shifted by an Einstein solution. The invariant information is the injection, cofiber, induced ranks, radicals, and extension class. A canonical branch split requires more structure than composition of differential operators.

On the conformal cylinder a different reduction occurs. In a selected derived zero-charge sector, all fifteen residual conformal generators are implemented by a BFV/Koszul receiver, and the vacuum and selected residual one-particle cohomologies are acyclic. The first surviving H^4 classes are deformation or vertex classes, not one-particle gravitons. This is a theorem about that selected compact sector. It is not a universal removal of radiative gravitons and it supplies no asymptotic particle interpretation.

The two laboratories therefore teach complementary lessons: additional directions can survive a compact local-gauge quotient, while a stronger zero-charge residual reduction can remove selected one-particle cohomology. Neither statement transfers to the other without a map.

5 Which directions survive nonlinear integrability?

Linearization stability on compact Cauchy surfaces has a classical history from Brill–Deser through Fischer–Marsden, Moncrief, and the Arms–Marsden–Moncrief momentum-map normal form [21, 22, 23, 24]. The general idea that a symmetric solution has a quadratic momentum-map cone is therefore not new. The contribution here is an exhaustive model-specific calculation for a fourth-order Weyl–Maxwell mode space and a separation of formal secular solvability from bounded solvability.

Let L be the linearized equation and $D^2E[u, u]$ the quadratic source. Two correction spaces are kept distinct:

$$\mathscr{X}_{\mathrm{ep}} = \left\{\text{finite sums of } t^p e^{-i\Omega t} \text{ with spatially periodic finite harmonic support}\right\}, \tag{20}$$

$$\mathscr{X}_{\mathrm{qp}} = \left\{\text{finite bounded quasiperiodic sums } e^{-i\Omega t}\right\}. \tag{21}$$

The first allows secular polynomials; the second does not.

Result card B: finite-harmonic second-order cone

Theory. Compact Weyl–Maxwell theory, with the Einstein–Maxwell comparison used to label branches.

Background. $\mathbb{R} \times S^1 \times S^2$ with the fixed magnetic bundle.

Domain. The complete declared finite harmonic space: generic Einstein and additional primaries, both parities, exceptional dipoles, twists, homogeneous generalized modes, electric flux, and Wilson directions.

Obstruction covectors. The five stabilizer covectors generated by time translation H, circle translation P_x, and rotations J_1, J_2, J_3.

Correction class. Finite exponential-polynomial corrections $\mathscr{X}_{\text{ep}}$.

Statement.

$$\mathscr{Z}_2^{\text{ep}} = \{u : \mu_H(u) = \mu_{P_x}(u) = \mu_{J_1}(u) = \mu_{J_2}(u) = \mu_{J_3}(u) = 0\}.$$

The five moment maps are necessary and sufficient in this correction class.

Not claimed. This is not a theorem for all smooth or Sobolev fields, retarded solutions, bounded dynamics, all-order integration, or stability.

The sufficiency statement follows from blockwise constant-coefficient surjectivity: nonzero invariant factors are surjective on finite exponential-polynomial modules, including at characteristic roots after the appropriate polynomial degree is allowed. The only zero invariant factors are the five stabilizer covectors. This is the explicit exhaustive content beyond the general momentum-map normal form.

Bounded continuation has a larger obstruction ledger. If the quadratic source is expanded by frequency shell and polynomial degree, then a bounded primitive exists only when three types of functional vanish:

$$\{\mu_X\} \ \cup \ \{\mathcal{P}_{j,r}\} \ \cup \ \{\mathcal{R}_{j,a}\}. \tag{22}$$

Here μ_X are the global zero-block Taub constraints, $\mathcal{P}_{j,r}$ are coefficients that would require positive powers of t, and $\mathcal{R}_{j,a}$ are adjoint-kernel pairings on nonzero characteristic shells. Their common nonlinear zero locus is not fully classified for arbitrary finite support. It is classified on several important subspaces. In particular, on the complete global plus $(\ell = 2, k = 0)$ additional-primary space, the bounded cone reduces to the standard global generalized-zero cone: no nonzero additional-primary amplitude survives. At other momenta, balanced sheets can remain.

The distinction is physical. A resonant source can have a formal primitive $te^{-i\omega t}$ and hence define a second-order jet, while failing every bounded quasiperiodic test. The programme therefore does not report "second-order solvable" without naming the correction class.

The same calculation also blocks a common branchwise simplification. Balanced Einstein–additional configurations can have zero total moment map while their projected Einstein and additional pieces carry equal and opposite nonzero charges. Consequently the ambient exact sequence does not descend to separately neutral branch fibers. Nonlinear reduction is intrinsically mixed.

6 Which formal asymptotic classes retain the extra solutions?

Compact charge constraints and black-hole endpoint conditions test different questions. The Schwarzschild analysis uses the Ricci perturbation $\psi_{\mu\nu} = \delta R_{\mu\nu}[h]$ as a second-order curvature variable in the fourth-order Bach system. On a Ricci-flat background the universal linear identity is

$$\delta B_{\mu\nu} = \frac{1}{2}\Box\delta R_{\mu\nu} + C_{\mu\rho\nu\sigma}\delta R^{\rho\sigma} - \frac{1}{6}\nabla_\mu\nabla_\nu\delta R - \frac{1}{12}g_{\mu\nu}\Box\delta R. \tag{23}$$

The Einstein image lies in $\ker\delta R$; the additional curvature content is visible in ψ.

Horizon analyticity does not force ψ to vanish. Regular ingoing additional curvature solutions exist in the declared real-frequency axial analysis. This already shows that future-horizon regularity alone does not select the Einstein image. It does not show that every local differential initial or boundary condition fails: imposing ψ and its normal derivative to vanish on a Cauchy surface is a local restriction that selects the Einstein image at linear order.

The infinity calculation instead gives a filtration-sensitive Phase-2 result. Introduce the dimensionless variables

$$\widehat{r} = \frac{r}{M}, \qquad \widehat{\omega} = M\omega, \qquad \Lambda = \ell(\ell+1), \tag{24}$$

and write formal radial powers in $\widehat{r}$. In ingoing Eddington–Finkelstein coordinates $(v, r, \vartheta, \varphi)$ let $F^v[u_1, u_2](r)$ denote the sphere integral of the v-component of the action-derived Lee–Wald current for two formal mode representatives. For a conjugate-frequency pair define the truncated radial slice form

$$\mathcal{N}_R[u_1, u_2] = i\int_{2M}^{R} F^v[\bar{u}_1, u_2](r)\,\mathrm{d}r, \qquad \mathcal{N}_\infty[u_1, u_2] = \lim_{R\to\infty}\mathcal{N}_R[u_1, u_2], \tag{25}$$

whenever the improper integral exists in the declared formal asymptotic class. "Finite radial slice form" below means finiteness of this quadratic form. It is neither an L^2 norm on monochromatic scattering states nor a wave-packet Hilbert norm, and the polar zero-sector quadratic form can vanish.

The dangerous-layer rule in this class is that a term $\widehat{r}^p(\log\widehat{r})^q$ is nonintegrable for $p \geq -1$ unless its invariant coefficient vanishes. Correctly retaining differentiated reconstruction forcing changes the old fixture-level conclusion.

Result card C: complete axial pilot and generic-angular polar disposition

Theory. Strict pure C^2 gravity.

Background. Schwarzschild with $M > 0$ in ingoing Eddington–Finkelstein coordinates.

Domain. Axial $\ell = 2$, $M = 1$, real $\omega \in [1/2, 3/4]$ for the complete all-row reconstruction; integer $\ell \geq 2$ and real $\widehat{\omega} \neq 0$ for the separate polar filtration; formal polyhomogeneous infinity data and the fixed action-derived pure-Weyl Lee–Wald representative.

Gauge and quotient. Regge–Wheeler/Zerilli-type parity reduction and metric reconstruction in the certified formal radial modules.

Statement. In the axial pilot the omitted $v\varphi$ Ricci row is the propagated constraint $C' = -2C/r$. Its zero fibre is a six-dimensional formal endpoint module: four Ricci-carrier lifts plus two complete Einstein-kernel columns. The Phase-2 X_0 is repaired by an exact transverse addition, whereas the legacy E_0 is not a complete Einstein solution. The repaired X_0 has no nonintegrable self coefficient and remains finite under the true rate-zero Einstein shear. Its cross pairing with the true oscillatory Einstein shear instead diverges. Thus the pilot contains a finite formal non-Einstein direction but proves no unrestricted representative-independent quotient statement. In the polar oscillatory block, the leading $p = 0$ coefficient matrix of the ordered current on (E, X_0, X_1, X_2) has generic rank three. Its one-dimensional filtered radical survives through the $p = 0, -1$ layers as a mixed Einstein/additional line. The canonical-pivot wall is empty on the physical domain. Define the specialized polynomial

$$R_\ell(x) := Q_{21}\big(\ell(\ell+1), x\big), \qquad x = \widehat{\omega}^2.$$

Away from $R_\ell(\widehat{\omega}^2) = 0$, the first finite $p = -2$ induced pairing is nonzero, so the mixed line is nonradical in this filtered sense. On $R_\ell(\widehat{\omega}^2) = 0$ that induced coefficient vanishes and the deeper filtration is open. Thus $Q_{21} = 0$ is not a second pivot or construction wall: it is precisely the degeneracy locus of the $p = -2$ finite-line pairing. Here Q_{21} is the certified bidegree-$(21, 21)$ polynomial. Exact Sturm counts give respectively $0, 6, 2, 6, 2$ real nonzero zeros of $R_\ell(\widehat{\omega}^2)$ for $\ell = 2$, $\ell = 3$, $4 \leq \ell \leq 10$, $11 \leq \ell \leq 40$, and $\ell \geq 41$; these are not physical scattering thresholds.

Not claimed. Generic-ℓ complete axial reconstruction, the axial X_2 branch, terminal-only polar prefixes, the deeper filtration on $R_\ell(\widehat{\omega}^2) = 0$, all-order polar solutions, and horizon-to-infinity matching remain open. No asymptotically flat phase space *with global matching*, Hilbert particle norm, scattering theory, quasinormal spectrum, ringdown, stability, particle, positivity, or quantum theorem follows.

Phase-3A endpoint completion: exact axial wave-packet flux

Theory. Strict pure C^2 gravity.

Background. Schwarzschild with $M = 1$.

Domain. Axial $\ell = 2$, positive-frequency support $\omega \in (1/2, 3/4)$, smooth compactly supported amplitude core and its $L^2([1/2, 3/4]; \mathbb{C}^3)$ completion; negative frequencies are fixed by the real-field involution.

Boundary spaces. The exact matching traces are

$$\mathcal{X}_{\mathscr{I}^-} = \operatorname{span}\{\mathrm{XI0}, \mathrm{XI1}, \mathrm{EI0}\}, \qquad \mathcal{X}_{\mathscr{I}^+} = \operatorname{span}\{\mathrm{XI2}, \mathrm{XI3}, \mathrm{EI2}\}.$$

The frozen action-derived current is pulled back through the complete metric reconstruction, including differentiated terms, and uses the outward Stokes orientation separately at each null end.

Statement. Both endpoint Grams have rank three and zero radical. After division by $\pi\alpha_{\mathrm{W}}$,

$$\det G_- = \frac{14155776}{125}\omega^3, \qquad \det G_+ = \frac{3538944}{125}\omega.$$

They have no frequency wall on the closed pilot interval and

$$\operatorname{inertia}(G_-) = \operatorname{inertia}(G_+) = (1, 2, 0) \qquad (\alpha_{\mathrm{W}} > 0).$$

The completion uses the auxiliary positive coefficient L^2 norm. Exact Sturm-certified Frobenius bounds give

$$\|a\|_{L^2} \leq \|G_\pm a\|_{L^2} \leq 645\|a\|_{L^2},$$

so the indefinite endpoint forms are uniformly nondegenerate. The past-boundary sign is already included in G_-; a global map would have to satisfy $T^\dagger J_{\mathrm{out}} T = J_{\mathrm{in}}$. Stationary local finite-tangential-jet trace-only improvements have vanishing cut terms on the wave-packet core, but unrestricted radial/corner improvement invariance remains open.

Not claimed. This endpoint theorem alone supplies no horizon-to-infinity connection, scattering channel, positive energy, particle norm, CPT metric, pole exclusion, stability, or unrestricted improvement invariance. The one-sided incoming population theorem below is an independent global result.

The endpoint signature admits an exact Witt decomposition. At $\mathscr{I}^-$, the null Einstein column EI0 and XI0 span a nondegenerate hyperbolic plane, while

$$-\omega^2\mathrm{XI0} - 2i\omega\mathrm{XI1} + \mathrm{EI0}$$

is orthogonal to that plane and has negative norm $-384\omega^3/5$ after division by $\pi\alpha_{\mathrm{W}}$. At $\mathscr{I}^+$, the corresponding hyperbolic plane is spanned by EI2 and XI3; its negative orthogonal line is generated by

$$i\omega\mathrm{XI2} + 4(4\omega^2 - i\omega - 2)\mathrm{XI3},$$

with norm $-384\omega/5$. Hence the two negative endpoint directions split algebraically as

$$(1,2,0) = (1,1,0) \oplus (0,1,0).$$

This is a Witt decomposition, not yet a Jordan-chain theorem. The full six-dimensional axial module is dimensionally incompatible with a scalar Regge–Wheeler square. Exact cyclic elimination instead gives

$$0 \longrightarrow M_{\mathrm{RW}} \longrightarrow M_{A_4} \longrightarrow M_x \longrightarrow 0,$$

while the metric Einstein kernel is another M_{RW} factor. Under $Z = y/[r^2(r-2)]$, the quotient M_x has the real $\ell = 2$ spin-one Regge–Wheeler potential $6(r-2)/r^3$. The complete six-state system therefore has the certified triangular factors

$$M_{\mathrm{RW}}^{\mathrm{metric}}, \qquad M_{\mathrm{RW}}^{\mathrm{carrier}}, \qquad M_x^{\text{spin-one}}.$$

The natural gauge retains a nonzero extension term, so a direct $\ker L_{\mathrm{RW}}^2 \oplus \ker L_x$ splitting and a time-translation Jordan interpretation are not claimed.

The incoming quotient map fixes the factor attribution without requiring such a splitting:

$$\pi_x(\mathrm{XI0}) = 2, \qquad \pi_x(\mathrm{XI1}) = -2i\omega, \qquad \pi_x(\mathrm{EI0}) = 0.$$

With

$$R = \mathrm{XI0} - \frac{i}{\omega}\mathrm{XI1}, \qquad S = -\omega^2\mathrm{XI0} - 2i\omega\mathrm{XI1} + \frac{4}{3}\mathrm{EI0}, \qquad E = \mathrm{EI0},$$

the exact incoming Gram divided by $\pi\alpha_{\mathrm{W}}$ is

$$G_-|_{(R,S,E)} = \begin{pmatrix} 624/(5\omega) & 0 & 576\omega/5 \\ 0 & -384\omega^3/5 & 0 \\ 576\omega/5 & 0 & 0 \end{pmatrix}.$$

Thus $\ker \pi_x = \mathrm{span}\{E, R\}$, the metric/carrier spin-two extension, has inertia $(1,1,0)$; the orthogonal lift S of the spin-one quotient is strictly negative. The two incoming negative directions therefore have different structural origins. With the quotient normalized to one, its exact norm is $-32/(15\omega)$. After an Einstein shear, the canonical all-frequency majorant has weights

$$\frac{576}{5}\omega(|e|^2 + |r|^2) + \frac{32}{15\omega}|s|^2.$$

On the positive-frequency Fourier core this is the homogeneous $\dot{H}^{1/2} \oplus \dot{H}^{1/2} \oplus \dot{H}^{-1/2}$ topology, up to Fourier normalization and the real-field extension. Weighted integrability does not itself define a threshold value $s(0)$. The canonical $\mathrm{diag}(1,-1,-1)$ fundamental symmetry supplies a positive classical Krein majorant; it is not a BRST-compatible Mannheim C operator.

There is nevertheless a sharp conditional consequence of the endpoint inertia. If a globally matched gauge-reduced trace map $T_\omega : U_\omega \to \mathcal{X}_\omega$ has rank r, then

$$n_-\Big(T_\omega^\dagger G(\omega) T_\omega\Big) \geq r - 1.$$

Indeed, in a Hermitian space of inertia $(1, 2, 0)$, projection onto the one-dimensional positive spectral subspace is injective on every subspace whose restricted form is nonnegative. Such a subspace therefore has dimension at most one. Consequently a rank-two populated image already contains a negative endpoint-flux direction, and full rank forces the pulled-back inertia $(1, 2, 0)$. If the matched frame is continuous, rank at least two at one interior frequency yields a smooth compact-frequency horizon amplitude with strictly negative endpoint flux on a neighboring frequency interval. Pairwise disjoint smooth frequency supports then give an infinite-dimensional negative subspace. Consequently no finite set of global charge constraints, finite-dimensional residual quotient, or finite exceptional-mode deletion can remove that endpoint index. This remains a statement about endpoint flux, not total canonical energy, a quantum ghost, or a complete scattering matrix; the horizon contribution to the global conservation law remains separate.

The conditional rank gate is now activated at the incoming endpoint. In factor-adapted future-horizon and $\mathscr{I}^-$ frames, exact triangularity gives

$$\det T_-(\omega) = -\frac{(2\omega - i)(4\omega - i)^2}{4(\omega - i)} A_{\mathrm{in},2}(\omega)^2 A_{\mathrm{in},1}(\omega).$$

The spin-two and spin-one potentials are real and short range. For their unit horizon-ingoing Jost solutions,

$$|A_{\mathrm{in},s}|^2 - |A_{\mathrm{out},s}|^2 = 1, \qquad s = 1, 2,$$

so $T_-(\omega) \in GL(3, \mathbb{C})$ for every real $\omega > 0$; the symbolic endpoint maps and recurrence pivots have no positive-real collision, while $\omega = 0$ remains separate. On the original pilot interval,

$$|\det T_-(\omega)| \geq \sqrt{5/2}.$$

Congruence therefore preserves the incoming inertia $(1, 2, 0)$; on every compact positive-frequency interval its L^2 wave-packet completion has infinite positive and negative index. This proves globally populated indefinite incoming channels, not the rank of T_+, nonzero reflection, absence of unstable poles beyond the separated-mode result below, positive total energy, CPT obstruction, or a complete scattering theory. Since $\ker T_-(\omega) = 0$, there is no future-horizon-regular solution with vanishing incoming trace: purely outgoing real-frequency axial modes and real-axis incoming spectral singularities are excluded for every $\omega > 0$. With the repository phase $e^{+i\omega t}$, growth is in the lower half-plane. Positivity of both reduced Regge–Wheeler Schrödinger operators excludes lower-half-plane growing Evans zeros for the three diagonal factors. The exact filtration maps preserve the future-horizon-regular and pure-outgoing germ classes there, so successive quotient elimination proves that the complete non-split six-state axial $\ell = 2$ system has no growing separated mode. This is not a time-domain boundedness, decay, completeness, or full PDE stability theorem; damped upper-half-plane poles remain open.

The exact local metric/carrier extension has rank one, but a hoped-for spectral-derivative shortcut does not presently select the damped resonance structure. In a globally rational gauge, the trace residue at the horizon forces any coefficient of $\partial_\omega A_{\mathrm{RW}}$ to vanish. Moreover, at a simple zero the statement $c = qA'_{\mathrm{in},2} \bmod A_{\mathrm{in},2}$ with unspecified q is tautological because $[A'_{\mathrm{in},2}]$ is a unit. The invariant class $[c]$, and hence the local Smith type, still requires certified QNM and adjoint germs plus a convergent Fredholm pairing.

An independent exact Laurent-current calculation at the future horizon gives an outward Gram of rank three and inertia $(1, 2, 0)$ for every real $\omega > 0$. Its metric/carrier spin-two plane is hyperbolic, and its orthogonal unit spin-one quotient again has norm $-32/(15\omega)$. Thus the factor anatomy agrees at the incoming endpoint and future horizon. Neither orientation of the horizon Gram is semidefinite, so the Stokes identity

$$H_{\mathcal{H}^+} + T_+^\dagger G_+ T_+ - T_-^\dagger G_- T_- = 0$$

does not determine the rank of T_+ without the outgoing connection.

After separate normalization to either the signature form $J_\sigma = \operatorname{diag}(1,-1,-1)$ or the triangular null form J_0, the same identity would make $\mathsf{S} = (R, A)^T$ a J-isometric embedding and the reflection defect $J - R^\dagger J R = A^\dagger J A$ nondegenerate with inertia $(1, 2, 0)$. Its correctly normalized determinant is

$$|\det \widehat{T}_-|^{-2} = |A_{\mathrm{in},2}|^{-4} |A_{\mathrm{in},1}|^{-2};$$

the raw determinant needs the endpoint volume ratio. This remains a conditional algebraic theorem, not an activated scattering result, because the typed full T_+, its same-frame normalization, and the global Stokes defect are not yet certified.

The remaining time-domain population problem is already smaller than a full matrix estimate. In a weight-normalized triangular factor frame, boundedness of T_-^{-1} reduces to threshold, high-frequency, and limiting-absorption bounds on

$$\frac{b}{A_{\mathrm{in},2}^2}, \qquad \frac{d}{A_{\mathrm{in},2} A_{\mathrm{in},1}}, \qquad \frac{bd - A_{\mathrm{in},2} c}{A_{\mathrm{in},2}^2 A_{\mathrm{in},1}}.$$

The scalar diagonal inverses are already bounded by their Wronskian identities; the non-split extension ratios are not.

The full 282-term Q_{21} is machine readable at `black_hole_programme/phase2/general_l_polar_extendible_current_closure/current_artifacts/q21-finite-line-factor.json` (SHA-256 `9c08f3c59adfd2da973872a0d718ef97249fa91b690f63df40bdcac81cae7f62`).

The old $\ell = 2$, $\widehat{\omega} = 3/5$ divergence table omitted differentiated c' forcing and the independent $v\varphi$ row in the axial reconstruction; it is superseded. That point is not on the polar R_2 locus, but this does not restore its earlier Einstein-selection interpretation. Existing AdS/Euclidean selection arguments operate in different boundary classes [25, 26]. A physical asymptotically flat theorem still requires a completed global Bach phase space, the outgoing/horizon map, and a well-posed exterior problem. The scoped axial incoming trace is now globally populated by future-horizon-regular solutions, but the complete scattering map is not. This is a one-sided counterpart, not yet the full analogue, of the global Hilbert-space isomorphisms available for linearized Einstein scattering on Schwarzschild [4]. Flat-space conformal-higher-spin scattering separately admits conformal-gravity states beyond ordinary Einstein gravitons [5]; it does not construct a Schwarzschild Bach connection. Covariant phase-space and canonical-energy methods provide the conceptual comparison, but their global hypotheses are not asserted here [2, 3].

6.1 From formal radial pairing to physical boundary conditions

The result above classifies a fixed formal infinity module and one action-derived current representative. The separate endpoint theorem adds exact L^2 wave-packet traces and their flux Grams. The incoming trace has now been matched invertibly to future-horizon-regular data. A complete physical scattering theorem still requires the outgoing and future-horizon flux blocks, a global gauge-audited asymptotically flat phase space, and well-posedness of the exterior evolution. The first validated full-amplitude calculation closed its diagonal ranks and local lower solves on the required first cell but ended in SHORTFALL because fixed interval frames lost the shared ω-correlation needed for cumulative lower transport. The analytic triangular proof bypasses that substrate limitation for T_-, not for T_+.

7 Can a positive quantum state be constructed?

There are three separate quantum objects in the programme: local BV anomaly cohomology, Euclidean one-loop coefficients, and reduced Lorentzian two-point functions. They are not interchangeable.

In four dimensions a local Weyl breaking at ghost number one and form degree four has essential type-B and type-A representatives of the form

$$\mathcal{A}^{(1)} = \frac{1}{(4\pi)^2} \int_M \sqrt{g}\, \omega \big(c_{\mathrm{W}} C_{\mu\nu\rho\sigma} C^{\mu\nu\rho\sigma} - a_{\mathrm{W}} E_4 + b_{\mathrm{W}} \Box R \big)\, \mathrm{d}^4 x. \tag{26}$$

In the action normalization of Eq. (1) and the repository's $\Box R = 0$ finite-counterterm convention, the computed coordinates are

$$\boxed{c_{\mathrm{W}} = \frac{199}{30}, \qquad a_{\mathrm{W}} = \frac{87}{20}, \qquad b_{\mathrm{W}} = 0.} \tag{27}$$

The $\Box R$ coefficient is scheme dependent and may be shifted by a local counterterm. The nonzero C^2 and Euler representatives are the essential type-B and type-A classes, respectively, in the Deser–Schwimmer classification[38]. A one-loop divergence, beta function, trace anomaly, and BV master-equation breaking are related only after the gauge-fixed determinant, normalization, descent, and quotient maps have been supplied.

The numbers in Eq. (27) are established conformal-graviton coefficients, not new constants. With the common $(4\pi)^{-2}$ prefactor and the anomaly written as $cC^2 - aE_4$, the convention crosswalk is

$$(c, -a)_{\text{here}} = \left(\frac{199}{30}, -\frac{87}{20} \right) = (c, -a)_{\mathrm{FT}}, \tag{28}$$

after translating the Euler-sign convention of Fradkin and Tseytlin[37]. The contribution here is their reconstruction from the declared determinant complex, including zero-mode, Jacobian, and ghost bookkeeping, and the explicit map to the nontrivial class in $H^{1,4}(s \mid \mathrm{d})$.

The complete local determinant and heat-kernel ledger is:

sector	operator	statistics	Z exponent	c_i	a_i
metric TT, upper	$\Delta_{2,\perp}(4)$	bosonic	$-1/2$	37/24	31/72
Diff–Weyl scalar ghost	$\Delta_0(-4)'$	fermionic	$+1/2$	89/120	269/360
metric TT, lower	$\Delta_{2,\perp}(2)$	bosonic	$-1/2$	29/8	181/72
transverse Diff ghost	$\Delta_{1,\perp}(-3)'$	fermionic	$+1/2$	29/40	79/120
exact sum				199/30	87/20

The listed c_i and a_i are net signed contributions: the statistics sign and the displayed determinant exponent have already been applied. They are therefore not the unsigned heat-kernel coefficients of the individual operators. All four factors are second order. Their bundle ranks are $5, 1, 5, 3$; the primes delete five scalar and ten transverse conformal-Killing ghost zero modes. York/Hodge Jacobians and the nonminimal doublets are already included. The table lists a_i with the sign convention $c_i C^2 - a_i E_4$.

The map from this Euclidean spectral ledger to a gauge anomaly is a residue and cohomology map, not a comparison of names. If $I_i(d)$ are the dimensionally continued integrated carriers, then

$$s_d I_i(d) = (d-4) A_i^4 + s_d B_i(d) + \mathrm{d} C_i(d) + O((d-4)^2). \tag{29}$$

Multiplication by the one-loop pole, replacement of the Weyl parameter by the ghost ω, addition of the diffeomorphism completion, and reduction modulo s and d give

$$\Gamma^{(1)}_{\mathrm{div}}(d) \xrightarrow{s_d,\ \mathrm{Res}_{d=4}} \mathcal{W}^{(1)}_{\sigma} \xrightarrow{\sigma\mapsto\omega+\mathrm{Diff}} Z^{1,4}(s \mid \mathrm{d}) \longrightarrow H^{1,4}(s \mid \mathrm{d}). \tag{30}$$

The nonzero coordinates in Eq. (27) survive the last quotient. This is what promotes the effective-action Weyl anomaly to an obstruction to treating Weyl symmetry as an exact quantum gauge redundancy in the strict fixed-field-content theory. These coordinates are stable under similarity transformations within the declared local operator and regulator class. They are not thereby invariant under a genuinely different complex contour or measure prescription: such a redefinition changes the quantum theory and requires its own regulator and quantum-action-principle analysis before any coefficient comparison.

Result card D: strict and compensated local quantum theories

Strict theory. Fixed-field-content Diff $\ltimes$Weyl pure C^2 gravity.

Domain. Complete declared gauge-fixed local BV quotient on a regular Bach-locus chart, with Euclidean one-loop coefficient input.

Statement. The regulated breaking has $c_{\mathrm{W}} = 199/30$, $a_{\mathrm{W}} = 87/20$, and vanishing parity-odd and chosen type-D coordinates. Its essential ghost-number-one, form-degree-four Weyl components are nonzero. The strict local Euclidean one-loop QME is obstructed.

Extended theory. In the formal τ-adic Wess–Zumino BV algebra with $s\tau = \omega$, the same cocycle is exact and a local counterterm restores the Euclidean QME at one loop.

Not claimed. The extension changes the BV theory. It gives no unconditional all-loop theorem, regulator/QAP for an arbitrary changed action, Lorentzian QME, Hadamard state, positive physical Hilbert space, particles, scattering, or unitarity.

The compensator statement is an exact BV realization of the familiar dilaton/Wess–Zumino mechanism, whose broader structure is known [41]. Its value is not a claim that a compensator is a free repair. Indeed, the passive compensator leaves a gauge-invariant dressed trace in the classical causal complex. Repairing that obstruction led to the changed counterflow experiment in Section 8.

Separately, on one complete causal Berger BV complex the local Hadamard singular structure and causal commutator are available, but a full BRST-compatible covariance is not constructed because its stationary Cauchy carrier and real structure are incomplete. This is a missing-object obstruction, not a proof that no BRST-compatible Hadamard covariance exists; details are recorded in Appendix A. The reduced positive η result above does not define an adjoint on this full BV complex. Thus no positive full-BV state or one-particle Hilbert space has been constructed.

The immediate quantum literature includes conformal-gauge fixing and its additional ghosts, local BRST cohomology, and several alternative quantization prescriptions[39, 35]. Phase 1 neither reproduces all of that literature nor supersedes it. It contributes a content-addressed strict/extended local-BV calculation and records the precise point at which Lorentzian state construction remains absent.

8 What the counterflow candidate taught us

The strict anomaly and passive-compensator causal obstruction motivated a small changed-theory ladder. Passive trace repairs and a one-phase active clock failed combined causal, sign, and charge tests. Two phases with an auxiliary diagonal $U(1)$ allow neutral counterflow:

$$\dot{\chi} = 0, \qquad \dot{\psi} = \Omega, \qquad Q_{\mathrm{diag}} = 0, \qquad E_{\mathrm{rel}} = \frac{1}{2}\mu^2\Omega^2 > 0. \tag{31}$$

The relative phase can move while the diagonal compact gauge charge vanishes. Because its stress tensor is built from the dressed metric, it is quadratically visible in the dressed-trace direction.

For self-containment, the stationary family can be stated directly. Let σ_i be left-invariant one-forms on S^3 with

$$\mathrm{d}\sigma_1 = -\sigma_2 \wedge \sigma_3, \quad \mathrm{d}\sigma_2 = -\sigma_3 \wedge \sigma_1, \quad \mathrm{d}\sigma_3 = -\sigma_1 \wedge \sigma_2, \quad \int_{S^3} \sigma_1 \wedge \sigma_2 \wedge \sigma_3 = 16\pi^2. \tag{32}$$

In the gauge $\tau = 0$, $A = 0$, and constant χ, take

$$\mathrm{d}s^2 = -\mathrm{d}t^2 + a^2(\sigma_1^2 + \sigma_2^2) + c^2\sigma_3^2, \qquad \psi = \Omega t, \tag{33}$$

and define

$$q = \frac{c^2}{a^2}, \qquad x = a^{-2}, \qquad C = \mu^2\Omega^2. \tag{34}$$

The exact stationary equations for Eq. (8) have rank three and give

$$M_2 = \frac{2C(4q-1)}{3qx}, \qquad V_0 = -\frac{C(q^2-5q+1)}{3q}, \qquad \alpha_B = \frac{2C}{qx^2}. \tag{35}$$

The connected trace-principal healthy stationary component is

$$x > 0, \qquad C > 0, \qquad \frac{13-3\sqrt{17}}{4} < q < \frac{1}{4}. \tag{36}$$

The word "healthy" here has a specific homogeneous trace-block meaning, not positivity of the complete physical theory. The exact coefficients are

$$K_{\mathrm{tr}} = -\frac{C(4q-1)}{2qx} > 0, \qquad M_{\mathrm{tr}} = \frac{C(2q^2-13q+2)}{6q} < 0, \qquad \lambda_{\mathrm{tr}}^2 = -\frac{x(2q^2-13q+2)}{3(4q-1)} < 0. \tag{37}$$

Thus $q < 1/4$ gives a positive trace velocity coefficient, while $q > (13-3\sqrt{17})/4$ gives the negative trace mass coefficient and purely oscillatory homogeneous trace roots. The conditions $x > 0$ and $C > 0$ mean a positive spatial scale and positive relative-phase kinetic normalization times nonzero clock rate. A complete Green homotopy is certified at the selected point below, not throughout this whole stationary family. The selected causal point fixes $\mu^2 = 1$ and

$$q = \frac{9}{40}, \quad x = 1, \quad C = \frac{9}{16}, \quad \Omega = \frac{3}{4}, \quad \alpha_B = 5, \quad M_2 = -\frac{1}{6}, \quad V_0 = \frac{119}{1920}. \tag{38}$$

The free BV object is the tangent complex of the displayed action and gauge transformations. Write its ghost and field bundles as

$$\mathcal{F}^{-1} = TM \oplus \mathbf{1}_{\mathrm{W}} \oplus \mathbf{1}_{U(1)}, \qquad \mathcal{F}^0 = S^2T^*M \oplus \mathbf{1}_\tau \oplus \mathbf{1}_\psi \oplus \mathbf{1}_\chi \oplus T^*M_A. \tag{39}$$

The minimal linear complex is then

$$\Gamma(\mathcal{F}^{-1}) \xrightarrow{q_{-1}} \Gamma(\mathcal{F}^0) \xrightarrow{q_0} \Gamma(\mathcal{E}) \xrightarrow{q_1} \Gamma((\mathcal{F}^{-1})^* \otimes \Lambda^4 T^*M), \tag{40}$$

where q_{-1} is the linearized Diff–Weyl–$U(1)$ action, q_0 is the Euler–Lagrange Hessian, q_1 contains the Noether identities, and $\mathcal{E}$ is the dual equation bundle to the field bundle. The differential orders are also fixed by the action: q_{-1} is first order in the diffeomorphism and diagonal-$U(1)$ parameters and zeroth order in the Weyl parameter; q_0 is fourth order in the metric C^2 block, second order in the dressed Einstein and phase blocks, and algebraic in the auxiliary connection before elimination; q_1 has the corresponding formal-adjoint Noether orders. The cotangent BV rows and standard antighost–multiplier doublets complete this to a cyclic complex. The 70-row certificate is a harmonic-component implementation of Eq. (40); the invariant definition is the action and this graded complex.

Result card E: causal construction and physical nonselection

Theory. Equation (8) with the gauge transformations (9).

Background. The selected causal biaxial Berger solution with $\mu^2 > 0$ and homogeneous trace inequalities (37), together with its connected trace-principal healthy same-field stationary retuning family.

Domain. The free cyclic BV complex (40); at the selected point its diagonal-$U(1)$ summand is contractible, and the physical spectral reduction uses complete $SU(2)_L \times U(1)_R$ isotypical blocks.

Gauge and charge. Diagonal gauge charge, relative-phase charge Q_{rel}, time translation D, and the background-preserving generator $K = D - \Omega R_{\mathrm{rel}}$ are kept distinct.

Causal statement. Exact retarded and advanced homotopies, nilpotency, cyclicity, and support identities hold at the selected point, and the original dressed-trace cohomology class is removed.

Physical statement. The complete both-k, $j = \frac{1}{2}$ reduced block contains a Hamiltonian–Hopf quartet. The obstructing factor is

$$(40z^4 + 773z^2 + 3748)^2, \qquad 773^2 - 4(40)(3748) = -2151 < 0.$$

With perturbations proportional to e^{zt}, the substitution $y = z^2$ gives two nonreal conjugate y-roots, hence four z-roots $\{\pm(\alpha + i\beta), \pm(\alpha - i\beta)\}$ with $\alpha\beta \neq 0$; two grow exponentially. The quartet persists throughout the connected trace-healthy same-field stationary family. Therefore no member of that family is selected as a robust linearly stable candidate.

PT/CPT disposition. The complex quartet is an exact infeasibility certificate for a positive pseudo-Hermitian metric on this physical block; it lies in the broken-PT regime and cannot be rescued by redefining only the inner product.

Not claimed. This is not an all-isotype spectrum, a familywide Green-homotopy theorem, a full-PDE stability theorem, or a no-go over other field contents and gauge architectures.

The rejection criterion was fixed before the computation: a successor candidate had to be linearly stable in every physical sector. One exact unstable physical isotypical block is sufficient to reject it. It is not necessary to finish every higher-j block to establish this nonselection. Conversely, the finite block does not by itself describe the nonlinear evolution of arbitrary PDE data.

The persistence statement is exact rather than a numerical continuation from Eq. (38). Restoring $w = z^2/x$, the family characteristic divisor contains the quadratic factor

$$F_2(q, w) = 16q^2w^2 + 24q(3 - 2q^2)w + 32q^3 - 108q^2 + 81, \tag{41}$$

with discriminant

$$\mathrm{disc}_w\, F_2 = 256q^5(9q - 8) < 0 \qquad \text{on (36)}. \tag{42}$$

Thus the two w-roots remain a nonreal conjugate pair everywhere on the connected component; they cannot cross onto the negative real axis or pass through a multiple root. Since $x > 0$, their square roots remain a Hamiltonian–Hopf quartet. This no-crossing lemma is the familywide argument.

The quartet also fixes the disposition of this block under pseudo-Hermitian and PT/CPT-type reinterpretations[10]. Write the finite reduced block in first-order form $\dot{u} = Au$, so the characteristic roots z are the eigenvalues of A, and let $H = iA$ be the corresponding Schrödinger-form generator.

If a positive metric η satisfied the pseudo-Hermiticity relation $H^\dagger\eta = \eta H$, then $\eta^{1/2}H\eta^{-1/2}$ would be Hermitian, its spectrum would be real, and every characteristic root z would be purely imaginary. The certified quartet has $\alpha \neq 0$, so no such positive η exists on this block. In PT language the block is in the broken-PT regime: the obstruction is spectral, not a choice of norm or inner product, and no alternative quantization prescription acting only through the inner product can reopen it.

The equal-frequency Pais–Uhlenbeck mechanism occupies a different spectral boundary. There the degenerate generator develops Jordan structure and the PT programme interprets the associated zero-norm states through a specially constructed C operator[12, 11]. Secular logarithms and Jordan chains also appear in our characteristic-shell reconstructions, so they are a useful classical analogy. The pilot has not constructed zero-norm decoupling, a C operator, or an equivalence with the equal-frequency oscillator. The analogy therefore locates where the Mannheim mechanism could live without promoting the logarithmic solutions to a quantum theorem.

The quartet is a spectral Hamiltonian instability associated with a Krein-sign collision in the standard Hamiltonian–Krein sense [9], not merely a coordinate drift. The global clock block has a different status. On the unrestricted phase space,

$$\Omega_{\rm rel} = \mathrm{d}Q_{\rm rel} \wedge \mathrm{d}\psi, \qquad H_{\rm rel} = \frac{Q_{\rm rel}^2}{2I} + E_{\rm geometry}, \tag{43}$$

so its secular dephasing is tangent to an action–angle family and passes an orbital, frequency-modulated interpretation. It does not cure the $j = \frac{1}{2}$ quartet.

The charge analysis gives an independent theorem. For phase clocks with linear charge constraint $CQ = q_0$ and clock velocity vector v,

$$D \text{ is null after reduction} \quad \Longleftrightarrow \quad v \in \operatorname{im} C^T, \tag{44}$$

while physical clocks are classes of v modulo $\operatorname{im} C^T$. In the selected model, fixing and quotienting $Q_{\rm rel}$ makes D null but contracts the relative-clock Darboux pair. Keeping the clock leaves $H_D = \Omega Q_{\rm rel}$ and raw D charged; $K = D - \Omega R_{\rm rel}$ preserves the background. No single declared charge fibre supplies both the proposed physical clock and a gauge time translation.

The counterflow result is therefore not a failed piece of engineering. It separates three notions that are often conflated:

$$\begin{aligned} \text{causal completeness} \quad &\not\Rightarrow \quad \text{positive reduced dynamics}, \\ \text{positive reduced dynamics} \quad &\not\Rightarrow \quad \text{acceptable clock/charge interpretation}. \end{aligned} \tag{45}$$

The causal theorem remains valid even though the candidate is rejected by a later physical gate.

9 Integrated Phase-1 verdict

The main findings can now be placed on the four levels of Eq. (2).

Level	Established in a declared scope	Still absent
Local linear systems	Complete causal BV complexes on selected compact backgrounds; exact additional compact-product and Schwarzschild curvature solutions	A single causal theory covering every laboratory and boundary class
Reduced classical directions	Additional axial/polar compact-product directions are nonradical; counterflow causal parent and exact charge classification; complete axial $\ell = 2$ Schwarzschild pilot with a finite fixed-representative extra direction but divergent oscillatory Einstein shear, polar finite mixed line with exact R_ℓ degeneracy locus, and exact axial endpoint L^2 flux Grams of rank three, zero radical, and inertia $(1, 2, 0)$; exact RW/RW/spin-one filtration and invertible incoming Schwarzschild map T_-; no growing separated axial $\ell = 2$ mode	Complete outgoing/horizon scattering map, time-domain PDE stability, positive energy theorem, and acceptable counterflow spectrum
Nonlinear continuation	Five-charge necessary-and-sufficient finite exponential-polynomial cone; bounded resonance ledger and classified subcones	Full bounded finite-support zero locus, retarded nonlinear existence, all-order integration, and stability
Quantum states/particles	Strict local one-loop BV obstruction; formal changed-theory Wess–Zumino resolution; structured positive η on selected reduced blocks; spectral PT no-go for the counterflow quartet	Genuine BRST-compatible C operator, full-BV Hadamard state, positive physical Hilbert space, particles, scattering, optical theorem, and unitarity

This supports the following central claim:

> **A pole-level diagnosis is too coarse, but the more complete reduction does not generically rescue the theory.** Some additional classical directions survive gauge and presymplectic reduction. Nonlinear moment maps, radial filtrations, local anomalies, inner-product compatibility, and reduced spectral tests then separate candidates for different reasons. The first changed theory to achieve a complete causal BV parent is rejected throughout its connected trace-healthy same-field family by a physical $j = \frac{1}{2}$ Hamiltonian–Hopf instability.

This is a classification ending, not a viable-theory claim and not a universal no-go. A different field content, a different gauge architecture, or a different admissible state prescription could define a new programme branch. The scoped Phase-2 update adds sharper comparisons without reopening or rewriting the frozen Phase-1 record.

10 Relation to prior work

The programme combines several established structures. Its claims should be read as explicit constructions and exhaustive calculations in specified models, not as inventions of the surrounding general theories.

Causal gauge complexes. Green-hyperbolic complexes, retarded/advanced Green homotopies, and their induced Poisson structures have a general Lorentzian theory [18]. Conformal deformation and Yang–Mills detour complexes place the Bach operator in a longer conformal-geometric lineage [19, 20]. The contribution here is the explicit action-derived conformal-gravity BV construction, its cyclic pairings and support identities, and transfer across the declared backgrounds.

Linearization stability. The relation between Killing symmetries, quadratic Taub constraints, and singular momentum-map zero sets is classical [21, 22, 23, 24]. The finite-harmonic theorem is an exhaustive Weyl–Maxwell realization: exactly five covectors exhaust the exponential-polynomial cokernel, exceptional and generalized modes are included, and bounded resonance conditions are separated from the formal cone.

Einstein selection and black holes. Boundary-condition selection of Einstein solutions in conformal gravity is well known in Euclidean and asymptotically (A)dS settings [25, 26, 27]. Prior Weyl-black-hole quasinormal analyses often isolate a second-order conformally transported sector rather than the full fourth-order Bach solution space [28]. The static spherical Bach-flat lineage begins with Riegert and Mannheim–Kazanas[29, 30]. Four-dimensional conformal gravity already has consistent AdS boundary conditions, response functions, and finite canonical charges[31, 32]; our open problem is specifically the asymptotically flat/null-infinity Bach phase space. Our complete axial $\ell = 2$ pilot shows that a finite formal non-Einstein representative survives while unrestricted Einstein shears need not preserve finiteness; the separate generic-angular polar result gives an exact mixed filtration with an exceptional-frequency wall. Neither provides a general asymptotically flat Bach phase space.

The magnetic product background belongs to the exceptional direct-product Einstein–Maxwell family introduced by Plebański and Hacyan [33]. The recent Weyl-gravity classification of separated $2 + 2$ direct products with electromagnetic and Yang–Mills sources proves generalized Birkhoff statements and organizes their Weyl-equivalence classes [34]. The present contribution is different: it fixes one compact magnetic bundle and computes its perturbative exact sequence, presymplectic form, nonlinear Taub cone, and resonances.

Quantum conformal gravity. One-loop divergences, conformal anomalies, BRST cohomology, and conformal gauge fixing have extensive prior literatures [36, 37, 35, 39, 40]. The phase-one quantum claim is narrower: an exact local-BV quotient and coefficient ledger for the declared strict theory, followed by a formal τ-adic Wess–Zumino trivialization in a changed theory. The Phase-2 comparison with pseudo-Hermitian quantum mechanics[13, 14] establishes a positive structured metric on selected reduced blocks and a spectral no-go on the broken-PT counterflow quartet, but no genuine BRST-compatible C operator or full Lorentzian state. The Hadamard distinction follows the microlocal state criterion[15]. Dirac–Pauli, fakeon, and conventional positive-state proposals are not settled.

Relational clocks. The construction of complete or partial observables from internal clocks is standard constrained-system territory[42]. The counterflow experiment adds a typed distinction between diagonal gauge charge, relative clock charge, time translation D, and the background-preserving combination K. No operational counterflow frequency ratio survived Phase 1, so the result is a charge/clock classification rather than an observational prediction.

11 Limitations and the next scientific boundary

The scoped Phase-2 results replace the earlier task list with sharper endpoint questions. The most important are not "finish quantum gravity." They are:

1. complete the asymptotically flat Bach phase space, classify the outgoing axial block T_+, horizon flux, and complex-frequency poles, and extend the global matching to the certified formal polar modules;
2. classify the axial X_2 branch, terminal-only polar prefixes, and the deeper filtration at $R_\ell(\widehat{\omega}^2) = 0$;
3. construct independent P, T, C data and prove BRST and causal-covariance compatibility, rather than promoting finite-block η existence to a state theorem;
4. supply the missing stationary Cauchy carrier and real structure, then decide the Hadamard covariance problem for the complete Berger BV complex;
5. if a new changed theory is proposed, require an open stationary locus with causal completeness, positive reduced dynamics, coherent charge interpretation, a preserved nonlinear sector, and at least one operational observable before calling it a candidate.

These are continuing Phase-2 gates. The bounded zero locus and mixed interaction class remain valid inputs, but are deprioritized as standalone interest levers. None is a condition for closing Phase 1. The stopping rule was outcome-independent: Phase 1 ended when the programme could give an exact viability classification of the first causally complete changed-theory candidate and reconcile it with the compact linear, nonlinear, black-hole, observer, and quantum results. That record is now frozen; the present additions are an explicitly layered update.

12 Computational accountability

The project treats computer algebra as an experimental instrument whose outputs require independent checks. Material results are represented by a producer, a machine-readable certificate, a verifier that does not merely rerun the producer, adversarial mutation tests where practical, a tiered test receipt, and a human-readable scope report. Exact algebraic ranks, Smith forms, cohomology witnesses, and characteristic factors use rational or algebraic arithmetic; numerical evidence is typed separately.

The authoritative Phase-1 record is

`PURE_WEYL_PROGRAMME_PHASE1_CLASSIFICATION_ENDING_V1`,

stored in

`reports/phase1-closure-claims-ledger-2026-07-22.json`.

The public repository root is

`https://github.com/area9innovation/weyl-gravity`,

with this manuscript and its machine-readable claim map under `https://github.com/area9innovation/weyl-gravity/tree/master/paper`. The JSON artifacts carry content hashes; no archival DOI is claimed at this stage. It binds each statement to a theory, background, phase space, correction or boundary class, lifecycle state, limitation, and source certificate. This paper has its own claim map and verifier; the live programme portal remains Paper 98. A claim that is not linked to evidence is left unpromoted rather than inferred from neighboring calculations.

Three aspects of this workflow matter scientifically. First, a failed promotion is retained: the invalid Berger round-sphere Hodge subspace, for example, was replaced by full $SU(2)_L \times U(1)_R$ isotypical blocks rather than silently repaired. Second, causal, spectral, algebraic, and quantum results carry different dependency labels. Third, stopping rules are fixed before the result is known. The counterflow candidate passed its causal gate and then failed a later physical gate; both facts remain part of the record.

13 Conclusion

The phase-one programme began with a common but underspecified question: whether the additional poles of pure Weyl gravity are physical ghosts. Its main conceptual result is that there is no single promotion from pole to particle. Local solutions, reduced classical directions, nonlinear tangent directions, and quantum states are different mathematical objects.

That distinction did not automatically rescue the theory. Additional compact-product directions survive local reduction, yet some fail bounded nonlinear continuation. Horizon regularity admits additional Schwarzschild curvature solutions. The complete axial $\ell = 2$ formal pilot supplies a finite non-Einstein representative but also a divergent cross term under the true oscillatory Einstein shear; the generic-angular polar filtration has a mixed finite line with an exact exceptional wall. On the axial pilot, exact three-dimensional wave-packet trace spaces at both null ends carry nondegenerate indefinite endpoint flux Grams, and the full incoming trace is globally populated at every positive real frequency by future-horizon-regular solutions through an invertible RW/RW/spin-one connection. One incoming negative direction lies in the spin-two extension and the other is the spin-one quotient line. Boundary dévissage excludes growing separated axial $\ell = 2$ modes, while the outgoing reflection block, damped quasinormal spectrum, and time-domain stability remain open. Selected compact blocks admit positive pseudo-Hermitian metrics, but neither a genuine BRST-compatible C operator nor a full Hadamard covariance follows. Strict pure Weyl gravity has a local Euclidean one-loop BV obstruction; a compensator removes it only in a changed theory. The first changed model with a complete causal BV parent is itself rejected by an exact family-persistent Hamiltonian–Hopf instability, and its complex spectrum excludes a positive-metric PT/CPT rescue.

The most defensible conclusion is therefore neither "the propagator already settled the physics" nor "constraints removed every ghost." It is:

> Pole-level signs are an initial diagnostic. Physical viability is a sequence of reductions and existence tests. In the laboratories updated through scoped Phase 2, different additional sectors survive different early tests, but no candidate reaches a positive full quantum state, and the first causally complete changed theory fails a robust linear physical-stability test. (46)

A Scoped nonpromotions retained in the evidence record

A.1 Dyonic stationary-family preflight

The attempted dyonic-flat-family extension fails before a sign calculation. For nonzero electric parameter the stationary symmetry has no continuous global Hamiltonian lift on the circle, since $\mathcal{L}_H A = E\,\mathrm{d}x$ changes the holonomy. Electromagnetic duality also fails to define a tangent direction in a fixed-Chern sector. Consequently that preflight constructs neither a fixed-coupling open stationary family nor a sign, current, inertia, or signature-wall theorem. It is retained to prevent an ill-typed background scan from being read as evidence for Result card A.

A.2 Berger Hadamard-covariance gate

The complete causal Berger BV complex has the required local Hadamard singular structure, causal commutator, and a conditional lifting statement. A full covariance is not constructed because the explicit 104-dimensional stationary carrier A_{104}, its Cauchy pairing $q_{104}^{\mathrm{Cauchy}}$, Green carrier $G_{104}^{\mathrm{Cauchy}}$, and real structure have not been supplied. The current linear ledger leaves 288 coordinates undetermined; the two tested compatible completions have zero-eigenspace dimensions 24 and 0. These figures diagnose missing input, not nonexistence. In particular, the finite reduced pseudo-Hermitian metric does not complete the full BV adjoint or a BRST-compatible Hadamard state.

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